

A Comparative Study of DenseNets for Vietnamese Traditional Music Genre Classification

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Abstract—The rapid progress of artificial intelligence has led to a corresponding increase in the demand for classification of music genres. Numerous firms have enlisted architects with exceptional expertise for the execution of this project. However, it is crucial to recognize that Vietnamese traditional music, along with other types of Asian traditional music, has not attained the same degree of outstanding performance qualities as other musical genres. This study aimed to assess the effectiveness of two architectural designs, namely Late Fusion Convolutional Neural Networks and Densely Connected Convolutional Networks, through the utilization of diverse visual transformations of musical patterns. The ongoing inquiry involved doing a comparative study on a meticulously preserved dataset of Vietnamese traditional music. The results obtained from our research inquiry have shown the architectural design that is best suited for this particular undertaking. The results of our analysis indicate that Late Fusion Convolutional Neural Networks are a better suitable option for achieving this specific goal. This study makes a substantial contribution to the field of music information retrieval (MIR) by investigating the effectiveness and precision of Densely Connected Convolutional Neural Networks (DenseNet)-based approaches in categorizing Vietnamese traditional music genres.

Index Terms—Music genre classification, Convolutional Neural Networks, Densely Connected Convolutional Neural Networks, Vietnamese traditional music, Music.

I. INTRODUCTION

Music genre classification plays a crucial role in music information retrieval (MIR) systems, facilitating effective process of browsing, searching, and recommending music [1]. Music genre categorization is essential in music information retrieval (MIR) systems since it enables efficient browsing, searching, and music recommendation processes. Due to the ability of discriminative feature extraction from audio spectrograms, Convolutional Neural Networks (CNNs) have become essential in the automated classification of music genres [2].

Nevertheless, the majority of prior studies have mostly concentrated on Western music genres, ignoring the challenges and opportunities inherent in Asian traditional music as a whole, with a particular focus on Vietnamese traditional music. Vietnamese music exhibits a wide range of instruments, styles, and regional variations, hence introducing additional complexities to the challenge of classifying genres. The objective of this study is to address the existing gap by examining the efficacy of various convolutional neural network (CNN)

structures in the classification of Vietnamese traditional music genres. Two potential architectures are compared in this study.

- Densely Connected Convolutional Networks (DenseNet) [3]: The densely connected model allows the reuse of features and the flow of information, which has an opportunity to improve the accuracy of categorization. The audio file will undergoes processing as Multi-Frequency Convolutional Arrays (MFCCs) and is then fed into DenseNet architecture to get the final outcome.
- Late Fusion Convolutional Neural Networks (Late fusion CNN) [4]: The architecture uses various spectrogram representations, namely Mel-Spectrogram (MLS), MFCC, and Short-time Fourier transform (STFT), to capture various features of the audio. The outputs of each group are combined for making predictions by merging these feature maps.

II. RELATED WORK

Vietnamese traditional music genre classification in particular and music genre classification in general are broadly researched fields in Music information retrieval [5] [6]. Numerous researches illustrate that CNNs, which is a class of deep learning models in various types of fields, including computer vision, and natural language processing, also have a superior role in Music genre classification [5] [6]. In addition, the CNNs model, known as a deep learning model, works well with images and can also process various types of musical transformations, such as short-term Fourier transform (STFT) [7] and mel spectrogram [8]. These transformation, similar to images, served as inputs for classification model. However, unlike normal images, these visual feature maps of musical pattern possess sequential features essential for the music classification task, which may be lost during CNNs training. Hence, G. Huang, Z. Liu, and L. Maaten devised a framework that employed Convolutional Neural Networks (CNNs) to construct DenseNet. Moreover DenseNet, a convolutional neural network, exhibits a high degree of interconnectivity among its feature maps. This methodology maintains the integrity of information by using feature reuse to maximize the network's potential, DenseNet provides condensed models that are very parameter-efficient and straightforward to train. Concatenating feature maps acquired by different layers enhances efficiency by increasing variety in the input of the following layers. On

the other hand, W. Seo and colleagues drew inspiration from DenseNet to construct a model specifically for the classification of music genres [4]. The late fusion technique is utilized in this model to combine three inputs obtained from a music file. Data is combined one stage ahead of the classification phase in late fusion. In particular, the CNN model receives input data with multimodal features in different feeds, and the output from each modality is concatenated. This method works especially well for capturing the unique features of each modality. The study demonstrated that their approach is highly appropriate for the objective of music genre classification. The architecture underwent further testing on multiple datasets, ranging from hundreds to thousands of songs.

III. METHODOLOGY

A. Overview

Two models were employed in this study: DenseNet and Late fusion CNN. DenseNet is a sequence of CNNs with dense connections, while Late fusion CNN, based on DenseNet, uses three distinct inputs in parallel. Late fusion is chosen for its state-of-the-art statistics when compared to other architecture in music genre classification task [4]. Therefore, DenseNet is perfect candidates for evaluated, because late fusion is inspired by DenseNet with addition modification. Furthermore, we conducted a comprehensive analysis of the methodology utilized by two models, focusing specifically on the Late Fusion Parallel approach.

B. Densely Connected Convolutional Networks

Let us begin by analysing the foundational principles of DenseNet. The DenseNet architecture is an advanced CNN architecture design that improves the capacity for feature reuse and representation augmentation. The DenseNet architecture facilitates the reception of input from all preceding levels within a block, in contrast to standard CNN which restrict input to the immediate preceding layer. The network's ability to acquire increasingly complex and diverse representations of incoming data is attributed to its robust and continuous information flow, which is facilitated by its highly interconnected topology. Illustrated in figure 1.

Started with a single image x_0 that is passed through a convolutional network, the follow formula [3, equation 2]:

$$x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}]) \quad (1)$$

shows that the l_{th} layer is presented with the feature maps of all previous layers $x_0, x_1, \dots, x_{l-1}$ as concatenated input inside a H_l composite function of three sequential operations: batch normalization (BN), rectified linear unit (ReLU) activation function, and a 3×3 convolution (Conv) [9]. Unlike ResNet [10] (the predecessor of Densenet), DenseNet employs concatenation to generate a continuous data flow instead of adding results from previous layers together. The model is a composite of many dense blocks, and the output feature maps become increasingly intricate as time progresses. In order to manipulate the output channels, transition layers have been incorporated between blocks to

perform convolution and pooling operations. The transition layers are composed of a batch normalization layer, a 1×1 convolutional layer, finally a 2×2 average pooling layer.

If each function H_l generates k feature-maps, it may be inferred that the l_{th} layer possesses $k_0 + k \times (l - 1)$ input feature-maps, where k_0 represents the number of channels in the input layer. One notable distinction between DenseNet and conventional network topologies lies in the ability of DenseNet to possess significantly narrower layers with the hyperparameter k denoted as the network's growth rate. An experiment [3] has proved that a rather modest growth rate is adequate for achieving state-of-the-art outcomes on the datasets that were subjected to testing. This phenomenon happened possibly because each layer possesses the ability to access all the feature-maps that precede it within its block, effectively granting it access to the network's "collective knowledge". The feature-maps can be regarded as the overall condition of the network. Each layer contributes k feature-maps to the current state. The growth rate governs the extent to which each layer provides novel knowledge to the global state.

C. Late Fusion Convolutional Neural Network

Late Fusion Convolutional Neural Network (Late fusion CNN) is inspired by DenseNet [3] [4]. The model utilizes all the previous output feature maps $0, 1, 2, \dots, l - 1$ as input feature maps, denoted as x_l . The mathematical expression for x_l is show as equation 1.

The Late fusion CNN model utilizes four dense blocks with transition layers in between to extract three independent characteristics of STFT [11], MFCC [12], and Mel-spectrogram (MLS) [8] in parallel [4]. STFT offers the benefit of comprehending the frequency flow inside a specified time interval while the signal frequency varies. Based on the result of the STFT, MLS's benefit is more sensitive to low frequencies than high frequencies, which is a human feature. The MFCC, which is derived from the MLS, is utilized to mitigate frequency correlations.

Initially, three inputs for this model pass 7×7 Convolutional layers with a stride of 2. Next, it pass through a 3×3 Max pooling layer with a stride of 2. Following that, there are dense blocks that is identical to the dense block used in DenseNet-121 [3], with the addition of a Dropout layer at the end of each layer. Next, a transition layer is added, which is identical to the transition layer used in DenseNet-121. The dense block is repeated four times, whereas the transition layer is repeated three times. Except for 4th dense block, there are 2×2 adaptive average pooling layers. The dense block consists of the following numbers of layers: 6, 12, 24, and 16, correspondingly. Ultimately, the three outputs from each group are combined and passed via a fully connected layer, followed by a SoftMax classification layer. Cross-entropy [13] is employed as the loss function. Model's architecture is illustrated by figure 2.

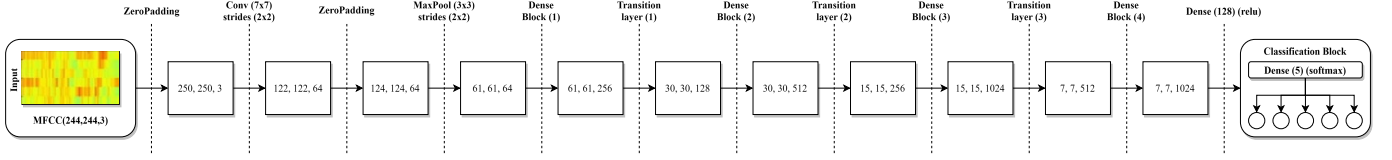


Fig. 1. DenseNet-121 [3, figure 2]

Compared to single-modal CNNs that rely on a single spectrogram, Late Fusion CNN captures different spectrograms distinctly, thus having access to a diversity of musical characteristics [4, table 5]. By combining them, the model leverages this complementary information. The MLS's perceptual focus might benefit from the time-frequency details of the STFT and the spectral envelope emphasis of MFCC. Moreover, by learning from multiple representations, the model potentially becomes more robust to variations in audio quality or recording conditions.

Specifically, when using STFT, MLS, and MFCC, the early fusion strategy may produce redundant information, given that the spectral features by MLS originate from STFT and those by MFCC originate from MLS. Consequently, the benefits of using multiple spectral features might disappear due to the overwhelming redundancy, which could potentially mislead the model [4]. The utilization of the late fusion technique has enabled this model to achieve a higher level of accuracy, surpassing its own previous performance [4, table 6].

IV. EXPERIMENT

A. Overview

The hardware configuration employed in the investigation was as follows: The 12th Generation Intel(R) Core(TM) i5-12400F offers a clock speed of 2500MHz, 6 cores, 12 logical processes, 16GB of RAM DDR4, and 15.8GB of storage. Regarding the graphic card, the NVIDIA GTX GeForce 1650 is utilized. The software utilized in this work encompassed Librosa for data preprocessing, spectrogram generation from audio files, as well as TensorFlow, Keras, and OS for model construction. The evaluations employed F1-Score and Accuracy metrics. We selected these indicators due to their efficacy in assessing models. As for F1-Score, the following formula

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

used two elements

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

which quantifies the ratio of accurate positive forecasts and

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

which measures the accuracy of properly predicting the proportion of real positive cases. F1-Score assessed the effectiveness of the model, particularly in datasets with skewed distributions, by integrating Precision and Recall to obtain a comprehensive perspective.

B. Dataset

1) *Overview*: This dataset, meticulously curated for the purpose of this research, encompasses a rich array of Vietnamese traditional music genres. Careful selection and assembly have resulted in a collection that showcases five distinct classes: "Xam", "Cheo", "Ca Tru", "Chau Van" and "Cai Luong." Each class comprises 500 audio files in the .wav format, providing ample diversity and depth for analysis. Sourced from Kaggle [14], the dataset's quality is deemed suitable for rigorous classification tasks. Each audio file is 30 seconds long, ensuring consistency across the entire dataset and facilitating seamless processing and analysis. This comprehensive dataset serves as a robust foundation for the exploration and classification of Vietnamese traditional music genres in our research endeavors.

2) *Preprocessing*: Each audio file was loaded with a sampling rate of 22050. The Mel-spectrogram was then computed from the loaded audio signal. The result was then converted to decibel units and resized to the desired shape of (244x244x3).

In advance of putting the DenseNet concept into practice, our preprocessing pipeline comprised of several critical stages. Initially, we established the foundation by creating functions dedicated to loading sample directories and processing audio files. These functions served to extract vital information from the dataset and transform the audio data into a format suitable for subsequent analysis. Upon loading the data, the subsequent step involved converting the raw audio data into MLS. This conversion was pivotal as it translates the temporal domain of audio signals into a frequency domain representation, making it more conducive for analysis by convolutional neural networks (CNNs) like DenseNet. Leveraging the capabilities of *Librosa* [15], a prominent Python library acclaimed for its prowess in music and audio analysis, we efficiently compute MLS, extracting meaningful features from the audio data. Once the MLS are computed, we organize them as images in a dedicated folder. This organizational strategy simplified data management and access, facilitating seamless integration with downstream processes such as model training and evaluation.

Subsequently, we proceeded to load the mel-spectrogram images previously computed and stored in the designated folder. Employing standard image loading techniques, we utilized the *OpenCV* library [16] to accomplish this task. After acquiring the MLS images, we prepared the data for model

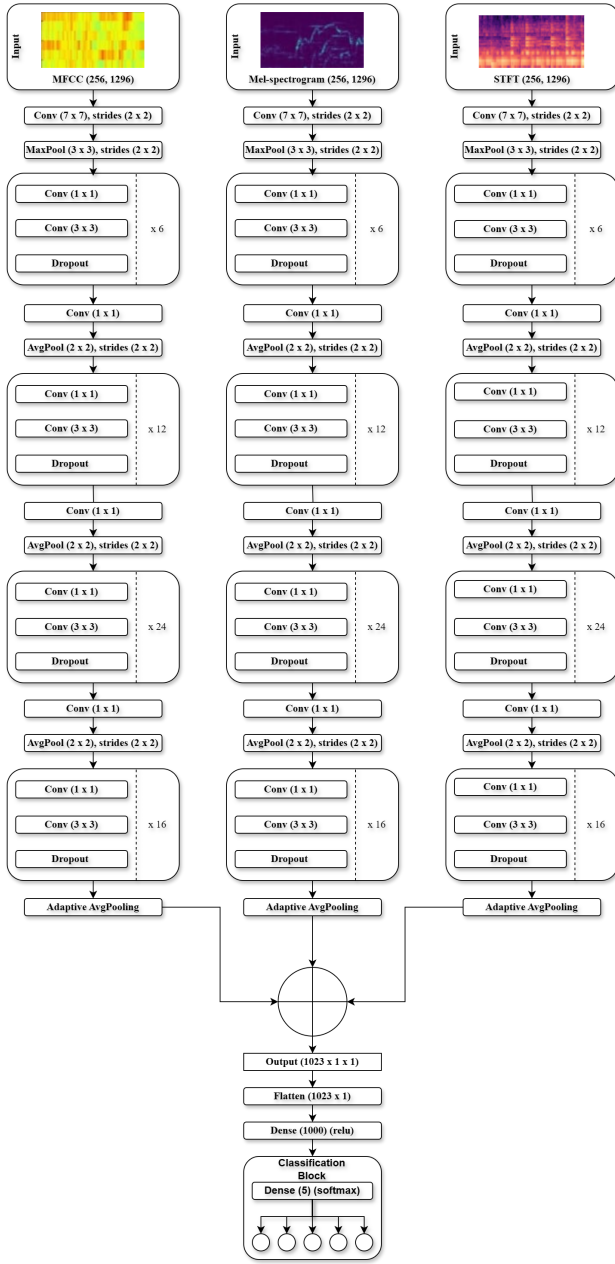


Fig. 2. Late fusion CNN [4, figure 1]

training by splitting the dataset into training, validation, and testing subsets. This partitioning was executed using the train test split function from the *sklearn* library [17], ensuring that each subset adequately represented the overall distribution of the data. Specifically, the dataset was randomly divided into a validation set consisting of 10% of the data, a test set comprising another 10%, and a training set containing the remaining 80%. To further streamline model training, we converted the data into *Numpy* arrays as float32. This conversion is imperative for maintaining numerical stability during training and ensures consistency in data representation across various platforms and libraries.

In preparation for the Late fusion CNN model, each input audio file was loaded and processed using three different types of audio transformations: STFT, MLS, and MFCC, utilizing the *Librosa* library [15]. Specifically, the audio files were computed for STFT (Sampling rate of 22115, Number of FFT bins is 510, Hop length of 512), MLS (Sampling rate of 12000, Number of FFT bins is 2048, Number of Mel bins is 256, Hop length of 256), and MFCC (Sampling rate of 22115, Number of FFT bins is 510, Number of Mel bins is 256, Hop length of 512). The result was three lists containing three types of transformed audio arrays, along with a label list, all processed using the *NumPy* library [18], with an output size of 256×1296 . Subsequently, to conform to the input format required by the model, we appended an additional dimension for each type of audio, with a size of one, through the reshape method of the *NumPy* library [18]. This is done to ensure that the input data for the model has the same dimensions as required by the format.

C. DenseNet

Throughout the refinement process of the DenseNet-121 model, iterative adjustments were made, leading to significant improvements from the initial configuration. Initially, the model utilized the base DenseNet-121 architecture from the library, with customized output adjustments for classifying 5 music genres, albeit encountering overfitting challenges during the preliminary trial. To counteract this, EarlyStopping was introduced with fine-tuned parameters. Subsequent enhancements were achieved by refining EarlyStopping to monitor validation loss and integrating Dropout, resulting in an enhanced model quality with an accuracy of 64%. In addition to model training, we adopted label binarization techniques to encode the categorical labels of our dataset. This step was crucial to ensure compatibility with the model's output layer and to streamline the computation of evaluation metrics. Specifically, the LabelBinarizer from the scikit-learn library [17] was employed to convert the categorical labels into one-hot encoded vectors. Furthermore, additional adjustments such as leveraging the Adam optimizer from the keras optimizers library and experimenting with various batch sizes were implemented. Ultimately, with further fine-tuning of parameters including batch size, epochs, and Dropout, the model achieved its peak performance, yielding an accuracy of $96.9\% \pm 0.7\%$ and an F1-score of $96.9\% \pm 0.7\%$.

D. Late fusion CNN

Firstly, we proceeded by partitioning three types of music lists and one label list into training, validation, and test sets. Similar to the data division method mentioned above, we also use the *sklearn* library and divide the data according to the same ratio for 3 types of audio transformations. Next, in re-architecting the model, we incorporated a dropout layer after each feature block and utilized early stopping to mitigate overfitting and halt the training process early if there was no significant improvement in minimizing the loss function on

TABLE I
STATISTIC OF MODEL DENSENET AND LATE FUSION CNN

	F1-score	Accuracy
DenseNet	96.9% $\pm$ 0.7%	96.9% $\pm$ 0.7%
Late fusion CNN	95.7% $\pm$ 2.9%	96.0% $\pm$ 2.7%

the validation set. We employed the LabelBinarizer from the sklearn library [17] to convert input data to array form and label encode the outputs for predictions

The model was compiled with a categorical cross-entropy loss function and the Adam optimizer, a popular optimizer in deep learning [19]. During training, the model was configured to save the best weights using the ModelCheckpoint callback, facilitating the retention of the model with the highest performance on the validation set. When evaluating the model's performance, multiple evaluation metrics were employed to assess its effectiveness in classifying music genres. Using the predictions generated by the model for the test dataset, we first converted the one-hot encoded predictions back to their corresponding labels. This allowed for a direct comparison between predicted and actual labels. After training and predicting with the model, it achieved an accuracy of 96.0% $\pm$ 2.7%, indicating the proportion of correctly classified instances. Additionally, the F1 Score, a harmonic mean of precision and recall, was computed at 95.7% $\pm$ 2.9%.

E. Comparison

After multiple iterations of training and testing, the efficacy of the DenseNet and Late fusion CNN models is demonstrated as below table: Both DenseNet and Late fusion CNN attained remarkable indexes of 96.9% $\pm$ 0.7% and 96.0% $\pm$ 2.7% respectively in Table I. We train repeat training process 10 different time and calculate average numbers. The use of late fusion has significantly improved the accuracy of the method, whilst the utilization of parallel techniques has led to greater efficiency. The hypothesis suggests that incorporating both spatial and temporal information concurrently results in a higher level of informative value as compared to only relying on temporal information.

F. Discussion

The Late-fusion model has an advantage over DenseNet when receiving up to 3 inputs, so the accuracy is also higher (experimental results show an increase in accuracy of about 1.7%), but the cost and calculation time are higher. It's also significantly more computationally intensive when having to process each flow in parallel. Our model training process also has some limitations. Due to the high density of both of the above models, the machine configuration was not able to meet it, so the accuracy was not as high as in theory. In the future, we will retrain the above two models with a machine with better configuration, to raise the accuracy to the state of the art level.

V. CONCLUSION

There is a growing interest in the domain of music genre classification. Architectural innovations have emerged in reaction to this expansion. We performed a comparative analysis of two DenseNet-based architectures using a dataset of Vietnamese traditional music. The outcome of the experiment revealed significant advancements in the classification of Asian traditional music, with a specific focus on Vietnamese traditional music. Furthermore, this study will contribute to Asian traditional music in general and Vietnamese traditional music in particular in music genre classification. In future endeavours, we will endeavour to expand the scope of this model to accommodate bigger datasets and include other traditional music genres.

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REFERENCES

- [1] M. Schedl, E. Gómez, and J. Urbano, "Music information retrieval: Recent developments and applications," *Foundations and Trends® in Information Retrieval*, vol. 8, no. 2-3, pp. 127–261, 2014. [Online]. Available: <http://dx.doi.org/10.1561/15000000042>
- [2] Y.-H. Cheng, P.-C. Chang, and C.-N. Kuo, "Convolutional neural networks approach for music genre classification," in *2020 International Symposium on Computer, Consumer and Control (IS3C)*, 2020, pp. 399–403.
- [3] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," 2018.
- [4] W. Seo, S.-H. Cho, P. Teisseire, and J. Lee, "A short survey and comparison of cnn-based music genre classification using multiple spectral features," *IEEE Access*, vol. 12, pp. 245–257, 2024.
- [5] K. Palanisamy, D. Singhania, and A. Yao, "Rethinking CNN models for audio classification," *CoRR*, vol. abs/2007.11154, 2020. [Online]. Available: <https://arxiv.org/abs/2007.11154>
- [6] Z. Nasrullah and Y. Zhao, "Music artist classification with convolutional recurrent neural networks," *CoRR*, vol. abs/1901.04555, 2019. [Online]. Available: <http://arxiv.org/abs/1901.04555>
- [7] T. Toshniwal, P. Tandon, and N. P., "Music genre recognition using short time fourier tranform and cnn," in *2022 International Conference on Computer Communication and Informatics (ICCCI)*, 2022, pp. 1–4.
- [8] H. Meng, T. Yan, F. Yuan, and H. Wei, "Speech emotion recognition from 3d log-mel spectrograms with deep learning network," *IEEE Access*, vol. PP, pp. 1–1, 08 2019.
- [9] K. O'Shea and R. Nash, "An introduction to convolutional neural networks," 2015.
- [10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015.
- [11] J. Allen, "Short term spectral analysis, synthesis, and modification by discrete fourier transform," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 25, no. 3, pp. 235–238, 1977.
- [12] B. Logan, "Mel frequency cepstral coefficients for music modeling," *Proc. 1st Int. Symposium Music Information Retrieval*, pp. 1–2, 10 2000.
- [13] A. Mao, M. Mohri, and Y. Zhong, "Cross-entropy loss functions: Theoretical analysis and applications," 2023.

- [14] H. M. Thang, "Vietnam traditional music (5 genres)," 2021. [Online]. Available: <https://www.kaggle.com/datasets/homata123/vntm-for-building-model-5-genres?rvi=1/version/5>
- [15] B. McFee, C. Raffel, D. Liang, D. P. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in python," in *Proceedings of the 14th python in science conference*, vol. 8, 2015.
- [16] Itseez, "Open source computer vision library," <https://github.com/itseez/opencv>, 2015.
- [17] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [18] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant, "Array programming with NumPy," *Nature*, vol. 585, no. 7825, pp. 357–362, Sep. 2020. [Online]. Available: <https://doi.org/10.1038/s41586-020-2649-2>
- [19] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2017.